
StormWeaver Studios

High Performance Graphics and Light Transport Research

CLASS REFERENCE

The sw-studios Document Class

A reference manual for the sw-studios document class

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This is an internal research document from StormWeaver Studios.

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1 Formal results: the rendering equation

This section exercises the proven-claim family (Definition, Theorem, Lemma, Proposition, Corollary, Proof) using a generic radiometry example, unrelated to any specific paper’s content — exactly what you’d reach for when writing up a proven result.

Definition 1.1: Energy-conserving BRDF

A BRDF $f_r(\omega_i, \omega_o)$ is energy-conserving if

$$\int_{\Omega} f_r(\omega_i, \omega_o) \cos \theta_o d\omega_o \leq 1 \quad \text{for all } \omega_i. \quad (1.1)$$

Theorem 1.1: Reciprocity of the ideal Lambertian BRDF

The ideal Lambertian BRDF $f_r = \rho/\pi$ satisfies Helmholtz reciprocity, $f_r(\omega_i, \omega_o) = f_r(\omega_o, \omega_i)$, for any albedo $\rho \in [0, 1]$.

Proof. The expression ρ/π does not depend on ω_i or ω_o individually, so swapping the two arguments leaves it unchanged. ■

Lemma 1.1: Fresnel reflectance bounds

For a dielectric interface with real index of refraction $\eta > 1$, the unpolarized Fresnel reflectance satisfies $F(\theta) \in [F(0), 1]$ over $\theta \in [0, \pi/2]$, monotonically increasing.

Corollary 1.1: Grazing-angle limit

As $\theta \rightarrow \pi/2$, $F(\theta) \rightarrow 1$ — every dielectric becomes a perfect mirror at grazing incidence.

Proposition 1.1: Diffuse albedo as a special case

Setting $\rho = 1$ in Definition 1.1 recovers a perfectly energy-conserving white Lambertian surface, saturating eq. (1.1) with equality.

Example 1.1: Fresnel reflectance of glass at normal incidence

For $\eta = 1.5$ (typical glass) and $\theta = 0$, the unpolarized Fresnel reflectance is $F(0) = \left(\frac{\eta-1}{\eta+1}\right)^2 = 0.04$ — about 4% of light reflects at normal incidence, matching the familiar rule of thumb for glass surfaces.

2 Draft-stage claims

This section exercises Conjecture and Remark — the standard academic slots for numerically-confirmed-only and open/discussion content, respectively.

Conjecture 2.1: Convergence rate of stratified MC estimators

Stratified sampling of the hemisphere reduces estimator variance by a factor of $O(1/N)$ relative to uniform sampling, for any BRDF with bounded second derivative.

Remark 2.1 (floating-point precision). *Accumulating radiance contributions in float32 across very long paths (30+ bounces) can lose precision for low-albedo scenes; switching the accumulator (not the per-bounce math) to float64 is usually sufficient and cheap.*

3 Implementation

This section exercises Algorithm, the three code-listing flavors, and Implementation Note — the practical, engineering-facing environments.

Algorithm 1: Persistent-thread wavefront scheduling (per SM)

Input: work queue Q of active path states

Output: updated queue after one wavefront pass

```

1 while  $Q$  is not empty do
2   pop a batch of paths from  $Q$ ;
3   evaluate next bounce for each path in the batch;
4   if path terminated (RR kill or escaped scene) then
5     write result to output buffer;
6   else
7     push updated path state back onto  $Q$ ;
  
```

BVH ray–AABB slab test

```

inline bool intersectAABB(const Ray& r, const AABB& box, float& tHit) {
    float tMin = 0.0f, tMax = r.tMax;
    for (int axis = 0; axis < 3; ++axis) {
        float invD = 1.0f / r.dir[axis];
        float t0 = (box.lo[axis] - r.origin[axis]) * invD;
        float t1 = (box.hi[axis] - r.origin[axis]) * invD;
        if (invD < 0.0f) std::swap(t0, t1);
        tMin = std::max(tMin, t0);
        tMax = std::min(tMax, t1);
        if (tMax <= tMin) return false;
    }
    tHit = tMin;
    return true;
}
  
```

Monte Carlo hemisphere integration

```

def estimate_irradiance(brdf_fn, n_samples=4096):
    """Uniform-hemisphere MC estimate of outgoing irradiance."""
    omega = sample_hemisphere_uniform(n_samples)
    cos_theta = omega[:, 2]
    pdf = 1.0 / (2.0 * np.pi)
    values = brdf_fn(omega) * cos_theta
    return np.mean(values / pdf)
  
```

Kernel launch configuration

```

dim3 block(128, 1, 1);
dim3 grid((numPaths + block.x - 1) / block.x, 1, 1);
wavefrontKernel<<<grid, block, 0, stream>>>(pathStates, numPaths);
  
```

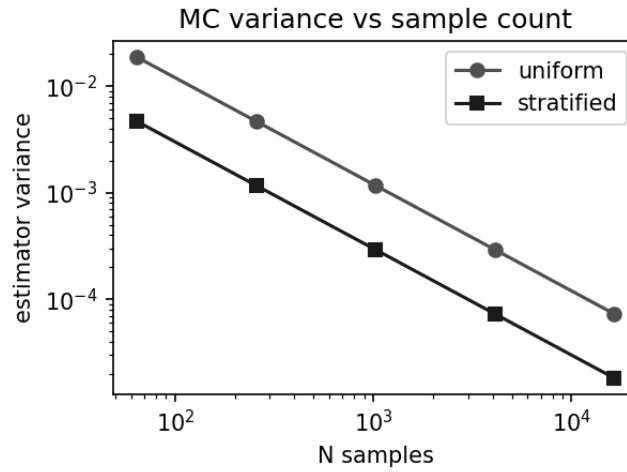
Implementation Note 3.1 (RNG seeding). Seed the per-path RNG with `path_id XOR frame_index`, not `path_id` alone — reusing the same seed every frame produces correlated noise patterns that show up as visible temporal flicker after denoising.

4 Tables, figures, and citations

This section exercises the table style, figure/caption support, and natbib citations together.

Table 1: Frame time (ms) at 1080p across denoiser configurations.

Configuration	Path tracing	Denoise	Total
No denoiser	18.2	—	18.2
Spatial only	18.2	1.4	19.6
Spatiotemporal	18.2	2.1	20.3

**Figure 1:** Estimator variance decreases with sample count N , consistent with the $O(1/N)$ rate proposed in Conjecture 2.1.

The reparameterization machinery used throughout this kind of derivation follows [LHJ19]; gradient estimator correctness follows the taxonomy of [ZSGJ21].

5 Environment gallery

A quick-reference catalog of every custom environment this class provides, each with minimal usage syntax. Come back here first if you forget how something works.

Definition 5.1: Gallery: Definition

`\begin{definition}{Name}{label} ... \end{definition}` — formal definitions, boxed, teal frame.

Theorem 5.1: Gallery: Theorem

`\begin{theorem}{Name}{label} ... \end{theorem}` — proven results, boxed, ink frame.

Lemma 5.1: Gallery: Lemma

`\begin{lemma}{Name}{label} ... \end{lemma}` — supporting results, boxed, lighter ink frame.

Corollary 5.1: Gallery: Corollary

`\begin{corollary}{Name}{label} ... \end{corollary}` — immediate consequences of a theorem or lemma.

Proposition 5.1: Gallery: Proposition

`\begin{proposition}{Name}{label} ... \end{proposition}` — standalone claims not depending on a preceding theorem.

Conjecture 5.1: Gallery: Conjecture

`\begin{conjecture}{Name}{label} ... \end{conjecture}` — numerically-confirmed-only, not yet closed-form. Teal frame, matches Definition.

Remark 5.1 (Gallery: Remark). `\begin{remark}[optional name] ... \end{remark}` — *unboxed, italic, for open questions and caveats.*

Example 5.1: Gallery: Example

`\begin{example}{Name}{label} ... \end{example}` — a worked numeric or concrete illustration. Gray frame.

Implementation	Note	5.1	(Gallery:	Implementation	Note).
<code>\begin{implnote}[optional label] ... \end{implnote}</code>	—	engineering/code	notes,		
	left-bar callout, not a math claim.				

proof: standard environment, plain style, ends with ■. Use for any theorem/lemma/proposition/corollary. algorithm (algorithm2e): `\KwIn, \KwOut, \eIf, \Return, \Comment*[r]{...}`.

Code: `cppcode`, `pycode`, and generic `swcode[title=...]{language}` — all via minted, ink/teal/ink title bars respectively.

Draft-only tags (for scratch docs, not finalized preprints): `proventag`, `numconfirmedtag`, `openitemtag` — no arguments, just wrap content.

Tables: `\swtablehead` after `\toprule` for a bold header row; `\swtablealt` before a long tabular for alternating row tints.

Figures: standard `graphicx/caption/subcaption`, styled captions (bold ink label, gray text).

Citations: `\cite{key}` / `\citep{key}` (default style: alpha, author-initial keys like [ZSGJ21] — avoid `\citet{}` in this mode) against `\swbibliography{refs}`.

Cross-references: `\cref{label}` / `\Cref{label}` auto-name the type; `\eqref{label}` for equations (section-scoped numbering throughout).

A Deferred proof: energy conservation of the Lambertian BRDF

This appendix demonstrates letter-numbered sections (section A instead of a numeric section) — the standard place for material moved out of the main narrative but still fully proven, matching the “harsh reviewer, nothing hand-waved” standard.

Proof. For $f_r = \rho/\pi$, the left side of eq. (1.1) evaluates to

$$\int_{\Omega} \frac{\rho}{\pi} \cos \theta_o d\omega_o = \frac{\rho}{\pi} \int_0^{2\pi} \int_0^{\pi/2} \cos \theta \sin \theta d\theta d\phi = \frac{\rho}{\pi} \cdot \pi = \rho \leq 1,$$

which holds for any $\rho \in [0, 1]$, proving Definition 1.1 directly. ■

References

- [LHJ19] Guillaume Loubet, Nicolas Holzschuch, and Wenzel Jakob. Reparameterizing discontinuous integrands for differentiable rendering. *ACM Transactions on Graphics*, 38(6), 2019.
- [ZSGJ21] Tizian Zeltner, Sébastien Speierer, Iliyan Georgiev, and Wenzel Jakob. Monte carlo estimators for differential light transport. *ACM Transactions on Graphics*, 40(4), 2021.